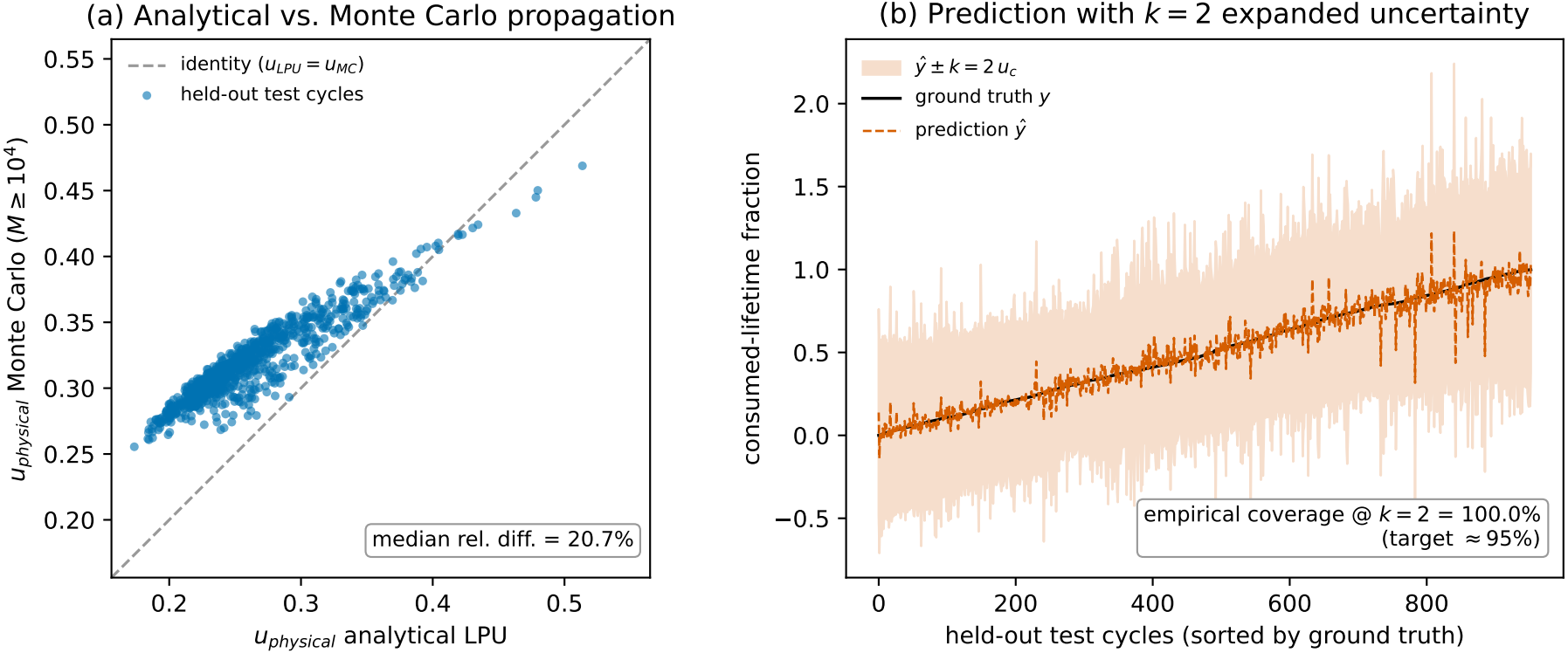

(a) Analytical vs. Monte Carlo propagation
identity ($u_{LPU} = u_{MC}$)
held-out test cycles
median rel. diff. = 20.7%
$u_{physical}$ Monte Carlo ($M \geq 10^4$)
$u_{physical}$ analytical LPU
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.2
0.3
0.4
0.5
(b) Prediction with $k = 2$ expanded uncertainty
$\hat{y} \pm k = 2\, u_c$
ground truth y
prediction $\hat{y}$
empirical coverage @ $k = 2$ = 100.0%
(target ≈ 95%)
consumed-lifetime fraction
held-out test cycles (sorted by ground truth)
2.0
1.5
1.0
0.5
0.0
−0.5
0
200
400
600
800